

🔥 XAI: Interpreting GAN Latent Spaces

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eXplainable
Artificial Intelligence

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

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<https://interpretable-ml-class.github.io/>

GANSpace: Discovering Interpretable GAN Controls

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Abstract

This paper describes a simple technique to analyze Generative Adversarial Networks (GANs) and create interpretable controls for image synthesis, such as change of viewpoint, aging, lighting, and time of day. We identify important latent directions based on Principal Component Analysis (PCA) applied either in latent space or feature space. Then, we show that a large number of interpretable controls can be defined by layer-wise perturbation along the principal directions. Moreover, we show that Pix2Pix can be controlled with layer-wise inputs in a StyleGAN-like

🍷 GANSpace: Motivation ---

- GANs are models trained to generate realistic-looking images well—and they do!
- They have not been designed, however, to make the image-generation process itself interpretable, or to support edits to generated images
- E.g.: This GAN generates faces well and can take in high-level styles or criteria (gender, age, race), but is not designed to facilitate post-hoc continuous and meaningful edits, such as head angle, hair length, lighting. . .



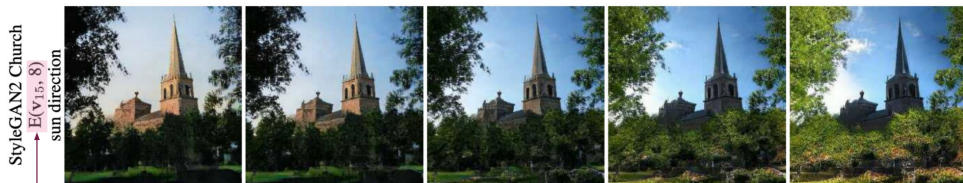
🍷 Main Contributions ---

- Applying a simple technique (**PCA**) in the latent space of GANs gives semantically significant directions.
- Perturbing inputs in these directions in certain layers of the GANs → interpretable edits in properties of generated images.
- Such edits are high-level and nuanced (e.g. object shape to background landscape)
- A user need only label each “control direction” once to edit the image

In plain english

In other words, the authors posit that exploring the “EiGANspace” gives us new controls to edit and possibly better understand GANs at low cost

🍷 Main contributions example ---



Perturbing the 15th principal component direction in the latent space, in the 8th layer.



Combining edits in various principal component directions to edit a generated photo.

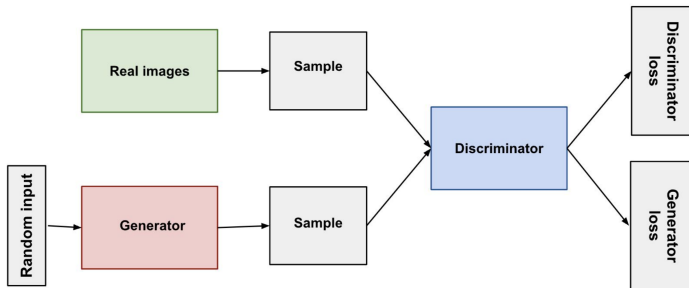


Background: Generative Adversarial Networks ---

Seminal Paper: Goodfellow et al. (2014)

- **Objective:** Generative Image Modeling—create realistic-looking, artificial images that resemble the training data.
- **GAN** = Generator (G) + Discriminator (D) networks, both CNNs in image context.
 - ▶ **Generator** G : maps latent $\mathbf{z}$ to image $G(\mathbf{z})$
 - ▶ **Discriminator** D : distinguishes real vs. generated images

Key Idea: “Back-and-Forth Between Generator and Discriminator”



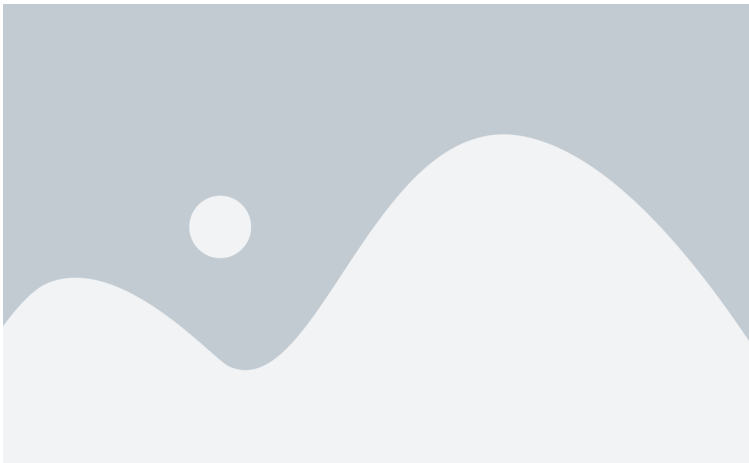
🍷 Background: BigGAN ---

- Class-conditional GAN for high-resolution ImageNet images
- Latent vector $\mathbf{z}$ is passed as **additional input** to all layers (no separate style vector)
- State-of-the-art for diverse, high-quality image generation



🍷 Background: StyleGAN ---

- Style-based generator: separates high-level attributes from stochastic variation
- Maps $\mathbf{z} \rightarrow \mathbf{w}$ via mapping network; $\mathbf{w}$ controls synthesis via AdaIN
- **w-space** is more disentangled than $\mathbf{z}$ -space



🍷 Background: PCA ---

- **Principal Component Analysis:** finds orthogonal axes of maximum variance
- Given data $\mathbf{y}_{1:N}$, PCA yields basis $\mathbf{V}$ and mean $\boldsymbol{\mu}$
- Project data: $\mathbf{x}_{1:N} = \mathbf{V}^T(\mathbf{y}_{1:N} - \boldsymbol{\mu})$
- First PCs capture the most variance; later PCs capture finer details

🍷 Related Work ---

- **Supervised directions**: require attribute labels to train classifiers in latent space
- **GAN analysis**: SeFa, closed-form factorization of generator weights
- GANSpace is **unsupervised** — no attribute annotations needed
- Related applications: camera motion, scene synthesis, image memorability

🍷 Methods: StyleGAN Approach ---

Idea: PCA directly on $\mathbf{w}$ -space samples

- 1 Sample N random $\mathbf{z}_{1:N}$, pass through mapping network to get $\mathbf{w}_{1:N}$
- 2 Apply PCA to $\mathbf{w}_{1:N} \rightarrow$ basis $\mathbf{V}$, mean μ
- 3 Represent each $\mathbf{w}$ in new basis: $\mathbf{x} = \mathbf{V}^T(\mathbf{w} - \mu)$

Editing

Move in the k -th principal direction with magnitude α :

$$\mathbf{w}' = \mathbf{w} + \alpha \mathbf{v}_k$$

Apply $\mathbf{w}'$ to some or all synthesis layers for global vs. localized edits.

🍷 Methods: BigGAN Approach ---

Issue: BigGAN has no separate style vector; $\mathbf{z}$ is passed as input at each layer



🍷 Methods: Granularity of Edits ---

Layerwise control

- **StyleGAN**: pass modified $\mathbf{w}'$ to some layers, original $\mathbf{w}$ to others
- **BigGAN**: pass modified $\mathbf{z}'$ to some layers
+ keep original $\mathbf{z}$ at start of network

Composed edits

- Edits can be *composed* by moving in **several** principal directions simultaneously
- Directions can be **labelled** by users to correspond to semantic concepts

🍷 Findings and Results ---



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🍷 Analyzing GANs Through Principal Components ---

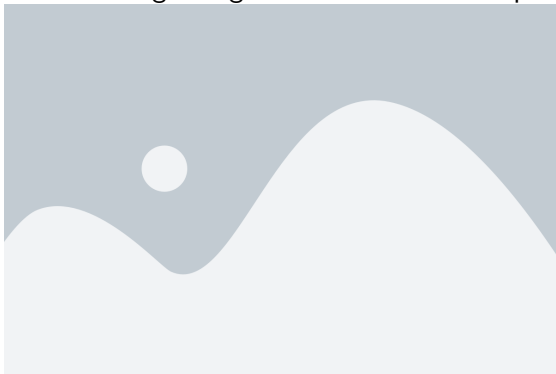
Limitations (inherited from dataset)

- StyleGAN faces cannot be translated in the image
- PCs for wrinkles/makeup have no effect on children/men



Entanglement/Superposition

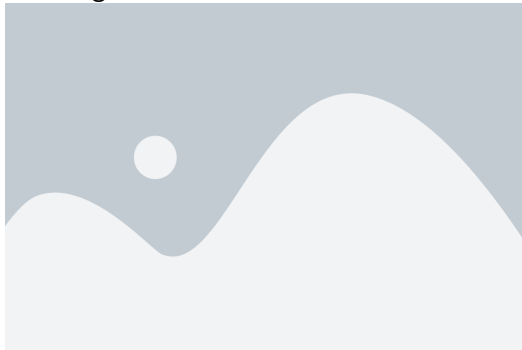
- One PC for StyleGAN cars corresponds to sportiness *and* open-road backgrounds
- Rotating a dog causes its mouth to open



🍷 Comparison with Related Techniques ---

GANSpace (PCA)

- Gives separated, ordered series of stylistic edit axes
- Random directions produce indiscernible changes



vs. Supervised methods

- Results similar to supervised approaches, without extra training
- Provides **many more** edit directions
- Some additional entanglement

🍷 Strengths / Weaknesses ---

Strengths

- Unsupervised and computationally simpler than supervised methods
- PC directions elucidate interpretable GAN latent representations
- After labelling directions, users can perform numerous edits on various styles

Weaknesses

- Approach limited to specific GAN architectures
- PCs are not *a priori* linked to semantic meaning
- Choice of which PCs to modify at which layers seems arbitrary
- Entanglement of PC styles hinders practicality
- Does this constitute actual **interpretability** for GANs?

🍷 GANSpace: Discussion Questions ---

- Are these principal components a useful tool for **interpreting** GANs?
- How does their utility compare with other interpretability methods we've studied?
- Most conceptually-aligned PCs for StyleGAN are in **w**-space; most useful PCs for BigGAN originate in the first linear layer (then projected to **z**-space)
- Do these findings provide novel insights into the models?
- How could this tool help **identify and reduce biases** in a model?

🍷 GANSpace: Summary ---

Experiment targets: **StyleGAN**, **BigGAN** (methods differed for the models)

- ① Changing the **first 20** PCs corresponds to changes in image **layout**, **configuration**, and **perspective**; later PCs dictate appearance and details
- ② StyleGAN's **first 100 PCs** sufficiently explain overall image appearance, distributed unimodally and almost independently
- ③ BigGAN's PCs seem to be **class-independent**

(Härkönen et al. 2020)

🍷 Paper 2: InterFaceGAN ---



InterFaceGAN: Roadmap ---

- Introduction
- Method
- Experiments

🍷 Background: What are GANs? ---

- **GAN**: generative models that generate realistic synthetic data mimicking training data
 - ▶ **Generator**: generates samples similar to training data
 - ▶ **Discriminator**: distinguishes generated (fake) from real samples
- **Latent representation**: a low-dimensional representation of an image
- **Latent space**: the set of all possible latent representations
 - ▶ Through adversarial training, the generator learns a mapping from latent space to real images



🍷 Background: GANs for Image Editing ---

- **Latent code:** the version of an image in the latent space
 - ▶ **GAN inversion:** identifying the latent code for a given image such that the generator could reconstruct it
- **Semantic editing:** editing the latent code to manipulate some features of the resulting image, such that only the desired features are changed





Related Work ---

- **Exploring latent spaces in GANs**

- ▶ Laine et al.: smoothly varying outputs across latent space
- ▶ Radford et al. observe the *vector arithmetic property*: `vector("King") - vector("Man") + vector("Woman") = vector("Queen")`
- ▶ Applications in camera motion, scene synthesis, memorability

- **Semantic face editing with GANs**

- ▶ Previous methods required carefully designed loss functions or specialized architectures
- ▶ No existing method to perform controlled facial editing by varying latent codes

Research Questions ---

- **How do semantics in the latent space originate, and how are they organized?**
 - ▶ Does there exist structure in the latent space relating to disentangled semantic representations?
- **What does a GAN actually learn with respect to the latent space?**
 - ▶ How does the GAN connect the latent space and the image semantic space?
- **How can the latent code be used for image editing?**
 - ▶ How are various semantic attributes of an individual's face (gender, age) determined and entangled?

🍷 Contributions and Main Findings ---

- Proposed **InterFaceGAN**, a framework to identify semantics encoded in the latent space of face synthesis models
 - ▶ GANs learn **latent subspaces** corresponding to specific attributes
- InterFaceGAN enables **semantic face editing with any pre-trained GAN**
 - ▶ Found theoretical and experimental results verifying that linear subspaces align with emerging semantics
- Applied InterFaceGAN to **real image editing**
 - ▶ Successfully edited attributes of real faces by varying the latent code

🍷 Method: Semantic Subspace ---

For any binary attribute, define the signed distance from hyperplane $\mathbf{n}$:

$$d(\mathbf{n}, \mathbf{z}) = \mathbf{n}^T \mathbf{z}$$

Key Hypothesis

The generator g satisfies:

$$f(g(\mathbf{z})) = \lambda d(\mathbf{n}, \mathbf{z})$$

where f scores the semantic attribute. The hyperplane $\mathbf{n}^T \mathbf{z} = 0$ separates the latent space into two semantic regions.

In practice, find $\mathbf{n}$ by training a **linear SVM** on labeled latent codes.

🍷 Method: Conditional Manipulation ---

To edit attribute A_1 while **preserving** attribute A_2 , project out the component of $\mathbf{n}_1$ along $\mathbf{n}_2$:

$$\mathbf{n}_1^* = \mathbf{n}_1 - (\mathbf{n}_1^T \mathbf{n}_2) \mathbf{n}_2$$



Move along $\mathbf{n}_1^*$ to change A_1 while remaining on the hyperplane of A_2 .

🍷 Experiment 1: Latent Space Separation ---

Model: PGGAN (Progressive Growing GAN) — trained on CelebA-HQ (30K images, 40 attributes)



🍷 Experiment 1: Hyperplane Assumption ---



Train 5 independent linear SVMs on pose, smile, age, gender, eyeglasses. Evaluate on validation set (6K samples) and full set (480K samples).

🍷 Experiment 1: Classification Accuracy ---



🍷 Experiment 1: Latent Space Visualization ---

Choose z far away from and on the decision boundary, then generate from it.



🍷 Experiment 2: Single Attribute Manipulation ---

Verify whether the semantics found by InterFaceGAN are manipulable.



🍷 Experiment 2: Distance Effect ---



🍷 Experiment 2: Artifacts Correction ---



Manually labelled 4K bad syntheses (with artifacts), trained a linear SVM to find the separation

🍷 Experiment 3: Conditional Manipulation ---

Study the disentanglement between different attributes.



🍷 Experiment 3: Conditional Manipulation Results ---



🍷 Experiment 4: Results on StyleGAN ---

StyleGAN extends PGGAN with:

Component	Description
Mapping Network	Casts $\mathbf{z}$ as $\mathbf{w}$ (“style vector”)
Synthesis Network	Uses style vector + noise via AdaIN
Bilinear Sampling	Replaces nearest-neighbor upsampling
Mixing Regularization	Uses 2 latent codes to generate



🍷 Experiment 4: W-Space vs Z-Space ---

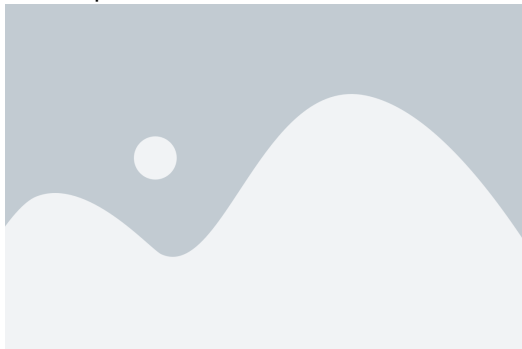


🍷 Experiment 5: Real Image Manipulation ---

Take a real face image $\rightarrow$ invert to latent code $\rightarrow$ use InterFaceGAN to edit.

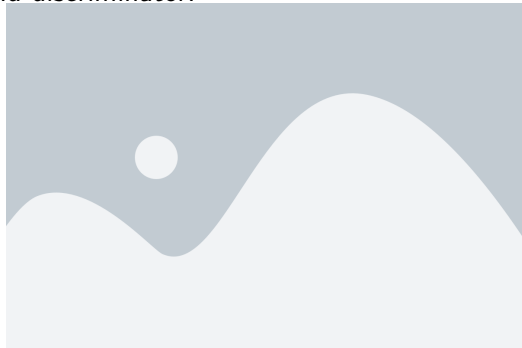
Latent Code Optimization

Optimize latent code with fixed generator to minimize pixel-wise reconstruction error.



Encoder-Based

Extra encoder learns inverse mapping from image to latent code by training with generator and discriminator.





InterFaceGAN: Discussion Questions ---

- 1 Do you believe the use of GANs for generating photo-realistic images should be **regulated**?
(Potential misuse)
- 2 Why do we think there are such simple linear boundaries separating binary semantics in the latent space (is it only for face GANs)?
- 3 How can analyzing semantic representations be used to **detect biases** in the training data?
- 4 Have you ever used GANs for photo generation, if so how realistic did you find them?
- 5 How relevant is this paper given that **GANs are progressively going out of style** (diffusion taking over)?

Thank You!

References --- |

Härkönen, Erik, Aaron Hertzmann, Jaakko Lehtinen, and Sylvain Paris. 2020. "GANSspace: Discovering Interpretable Gan Controls." In *Advances in Neural Information Processing Systems*.

Shen, Yujun, Jinjin Gu, Xiaoou Tang, and Bolei Zhou. 2020. "Interpreting the Latent Space of Gans for Semantic Face Editing." In *Proceedings of the Ieee/Cvf Conference on Computer Vision and Pattern Recognition*.